

Participant: 1, CS master student, on-site.

Background:

The participant knows Bayesian optimization and uses it frequently in his studies and works.

Homepage:

The participant understood the optimization concept.

Tutorial:

He was satisfied with the effect of the tutorial because he could clearly perceive what could be done with this service.

1st round

Specify page:

He first chose optimizing the colour board design as optimization scenario, the variables he set were colour contrast and saturation, but he found it hard to set the objectives as the objectives he wanted were too subjective, like clarity of information and satisfaction.

Optimize page:

He used the travel example given to start the optimization process in the end.

He thought that the measurement of the objective he input was not very accurate. For example, for the cost objective, when he saw a new solution, he found that the measurement of the cost he entered for the previous solution may be too high.

Result page:

At first, he found it difficult to understand why the option he had evaluated appeared on the result page. He thought the result page would propose a new solution which is more optimal than previous ones.

2nd round

Specify:

After getting familiar with using the service, he wanted to optimize the stock investment. The variables included expected income for the past three months, and the closing price of the day; Objectives: Minimize investment; Maximize the expected return.

Optimize:

After restarting the program, bugs occurred when the participant clicked "I want to evaluate this one", so the process didn't continue.

Findings:

- The participant thinks the objectives in the travel example are not completely accurate, because there may be a correlation between cost and satisfaction and they will affect each other.
- The participant thinks that weights can be added to objectives. For example, in the example of optimizing stocks proposed by him, when evaluating the solution, the weight of investment may be greater (for example, 80%), and the weight of return may be smaller (for example, 20%).

- The participant found it difficult to conceive of accurate, non-subjective objectives; and also he found it difficult to accurately reflect his measurement for those objectives in practice, as the measurement standard might change as new alternatives of solutions appear.